

EduMan

Manage Every Student

COMPLETE PRODUCT BLUEPRINT

A multi-institute Education Management Platform for tutors, tuition centres, coaching centres and educational organisations.

PRODUCT VISION • BRAND IDENTITY • FEATURES • WORKFLOWS • SECURITY • TECHNICAL REQUIREMENTS

1. Product overview

EduMan is an all-in-one Education Management Platform (ERP) designed to simplify daily operations for individual tutors, tuition centres, coaching centres, educational institutes and multi-centre organisations.

It combines student and parent management, attendance, fee records, examinations, timetables, staff management, communication, reports and AI assistance in one central platform.

The central objective is to make institute management simple for non-technical users while providing the tools required to operate efficiently.

EduMan is a multi-tenant platform: one owner can manage multiple independent institute workspaces from a single account.

Problem	EduMan solution
Scattered student details	Centralised student profiles and searchable records
Paper attendance registers	Digital attendance with a simple teacher workflow
Fees tracked in notebooks	Fee structures, payment records and outstanding balances
Manual parent reminders	Scheduled, editable AI-assisted SMS reminders
Parents lack timely updates	Parent portal, in-app notifications and SMS
Multiple centres managed separately	One owner account with isolated institute workspaces
Time-consuming reports	Automatic reports, analytics and AI summaries

2. Intended users

User group	Primary use
Individual/home tutors	Manage students, attendance, fee records, homework and parent communication.
Tuition/coaching centres	Manage batches, teachers, student records, schedules and fee plans.
Educational institutes	Manage academic sessions, staff, students, examinations and reports.
Multi-branch organisations	Manage multiple institutes and branches through one owner account.

3. Brand identity

Name, logo and tagline

Element	Specification
Brand name	EduMan
Meaning	Education Management
Tagline	Manage Every Student
Category	Education Management Platform / Education ERP
Logo	Registration-book icon above the app name, with the tagline beneath.
Style	Minimal, professional, recognisable and legible at small icon sizes.

Create stacked, horizontal and square-icon logo variants, plus light- and dark-background versions. Supply editable SVG master artwork and transparent PNG exports.

Primary palette — blue

Shade	Hex	Use
Lightest blue	#B3CCE0	Highlighted backgrounds and selected rows
Light blue	#8DADC6	Borders and secondary elements
Supporting blue	#668DAC	Charts and supporting icons
Main blue	#204E72	Brand colour, primary buttons and active controls
Deep blue	#113B5B	Hover states
Dark blue	#0E324D	Strong emphasis
Deepest blue	#0A283F	Deep accents

Secondary palette — green

Shade	Hex	Use
Lightest green	#B0C9AB	Success backgrounds
Light green	#8BA685	Secondary accents
Supporting green	#65835E	Charts and illustrations
Main green	#3F6038	Positive status indicators
Deep green	#36582F	Strong positive emphasis
Dark green	#2C4F25	Dark accents
Deepest green	#193D11	Dark-mode success

Light and dark themes

Element	Light mode	Dark mode
Main background	#FFFFFF	#111111 carbon black
Cards	#FFFFFF	#1A1A1A
Elevated panels	#F5F8FB	#222222
Borders	#D9E2EA	#333333
Main text	#0A283F	#F5F5F5
Secondary text	#0E324D	#B3CCE0
Primary button	#204E72 + white text	#B3CCE0 + #111111 text
Success background	#B0C9AB	#193D11
Success text	#193D11	#B0C9AB

Every user independently chooses light or dark mode. Use blue for branding and interactions, green for positive states, red for absent/failed/critical states, amber for pending/warning states, and neutral backgrounds for most surfaces.

Typography

Font	Usage
Calistoga Regular	Logo wordmark and selected brand headings
Inter Bold	Dashboard and section headings
Inter Medium	Navigation and buttons
Inter Regular	Body text, forms and tables
Space Mono Regular	Student IDs, organisation codes and reference numbers
Space Mono Bold	Emphasised identifiers

Use Inter as the main interface font. Include the supplied Calistoga and Space Mono files in the project assets where licensing permits.

4. Platforms and access

- Responsive website for desktop and mobile browsers.
- Installable Progressive Web App (PWA).
- Dedicated Android application.
- Downloadable Windows, macOS and Linux desktop applications.
- iOS may be supported as an additional client when ready.

All clients must use the same secure backend, accounts, permissions and data. The interface must adapt to mobile, tablet and desktop.

5. Multi-institute organisation model

- One owner account can create and manage multiple institute workspaces.
- Each institute receives an automatically generated unique organisation code.
- Each institute has its own name, logo, contact details, branches and academic sessions.
- Students, parents, staff, courses, batches, attendance, fees, timetables, announcements and documents are institute-specific.
- The owner can switch workspaces through a central institute selector.
- Cross-institute access is permitted only when explicitly granted.
- Data isolation must be enforced by the backend and database, not merely by hiding interface elements.

Example: one owner manages Institute A, Institute B and Institute C from one account. Each institute retains independent records and settings.

6. User roles and account rules

Role	Responsibilities
Super Admin	Manage the EduMan platform, organisations, users, plans, support and system settings.
Owner	Manage institutes, branches, staff, records, permissions and overall operations.
Admin	Manage authorised institute administration.
Manager	Handle delegated operational responsibilities.
Teacher	Manage assigned classes, attendance, homework, marks and permitted parent communication.
Parent	View linked children's information and communicate with the institute.
Accountant	Record external payments, manage fee records and view financial reports.
Receptionist	Manage enquiries and permitted student/contact records.
Custom roles	Use administrator-defined permissions.

- Students do not receive individual app login credentials; parents are the account holders.
- Parent accounts access information for linked student profiles.
- Only owners and authorised admins can create or edit student profiles.
- Parents can view permitted information but cannot edit student records.
- Teachers access only assigned students and batches, subject to permissions.
- Each role receives a fixed, simple dashboard.
- Every protected action requires server-side permission checks.

7. Dashboards and navigation

Dashboard	Main sections
Owner	Students, staff, institutes, branches, attendance, fees, exams, announcements, reports, institute switcher
Admin	Students, parents, teachers, attendance, fees, timetables, reports, communication, settings
Teacher	Assigned batches/students, attendance, timetable, homework, exams, marks, materials, parent queries
Parent	Linked children, attendance, fees, history, receipts, timetable, results, homework, materials, messages
Accountant	Fee records, external payment entry, outstanding balances, receipts and collection reports
Receptionist	Student directory, parent contacts, enquiries and permitted student actions

Use a bottom navigation bar for the main sections. Dashboards are fixed and simple; users cannot rearrange widgets. Tables support search, sorting, filtering, pagination and column customisation.

8. Student and parent management

- Create student profiles with organisation-defined custom fields.
- Automatically generate unique student IDs and organisation codes.
- Store parent details, contact information, course and batch assignments, academic history and relevant documents.
- Allow a student to belong to multiple courses and batches simultaneously.
- Import/export student and teacher records using Excel/CSV.
- Support organisation-customisable import templates and required columns.

- Create parent accounts and link them to student records.
- Send parent account setup credentials by SMS using secure credential-delivery practices.
- Parents can view attendance, fees, results, timetable, homework, study materials, announcements and permitted communication.

9. Attendance management

Default workflow: When a teacher opens the attendance register for an assigned batch, all students are initially marked Present. The teacher marks absent students, records late arrivals or early departures where applicable, reviews the register and finalises it.

- Manual attendance with all-present default and teacher marking absentees.
- Late arrivals and early departures.
- Student leave requests and authorised approvals.
- Permission-based access for owners, admins and teachers.
- Daily, weekly, monthly and yearly reports for individual students and batches.
- Attendance history and audit logs.
- Daily attendance summary for parents.
- Prepare attendance notifications after the teacher finalises the register.

The initial all-present state must be visible and reviewed by the teacher before finalisation. An unreviewed register must not silently become confirmed attendance.

10. Fee management

EduMan records fees and payments collected outside the app. It does not collect payments, process transactions or provide an in-app payment gateway.

- Course-wise, batch-wise, individual student and custom fee structures.
- Instalments, discounts, scholarships, fines and custom charges.
- Record cash, UPI, bank transfer and other externally collected payments.
- Support partial payments and calculate remaining balances.
- Calculate overdue amounts and outstanding fees.
- Maintain payment history and generate downloadable receipts.
- Generate monthly fee collection and pending-payment reports.
- Stop outstanding-fee reminders when the relevant amount is fully paid.

11. SMS and in-app communication

External automated messaging channel: SMS only. Do not integrate WhatsApp for automated sending. Relevant notifications and messages must also be available inside EduMan.

Message types

- Fee reminders; daily attendance notifications; parent announcements and circulars; examination result notifications; timetable changes; account setup messages; other authorised administrative notifications.

AI message editor and dynamic fields

AI prepares personalised drafts. Staff can edit the full text, insert dynamic fields, select recipients and review each recipient's preview.

Dynamic fields: {{StudentName}}, {{ParentName}}, {{InstituteName}}, {{PendingAmount}}, {{DueDate}}, {{AttendanceStatus}}, {{BatchName}} and other approved fields.

Recipient selection and sending

- Search and filter the student list.
- Select one student or multiple students.
- Resolve the linked parent and permitted contact number for each selected student.
- Generate personalised drafts and preview each recipient's final message.
- Allow authorised staff to edit and approve messages.

- Choose an approved sender arrangement supported by the SMS provider.
- Record every message and its delivery status.

Fee reminder schedule

- AI prepares the fee reminder one day before the due date.
- Include student name, parent name, outstanding amount, due date and institute details where available.
- Staff can edit and approve the draft.
- Institute settings determine whether each message requires individual approval or can use a previously approved template and schedule.
- Recalculate outstanding balances after external payments are recorded.
- Do not send an outstanding-fee reminder for a fully paid fee.

Attendance notification schedule

After the teacher reviews and finalises attendance, EduMan prepares the relevant parent message. It is sent by SMS and/or delivered in-app according to institute notification and approval settings.

Sender identity and provider constraints

The system should support authorised owners, admins and approved teachers initiating messages through an approved sender arrangement. SMS providers and telecom rules may require registered sender IDs, verified routes or approved device-based sending. Do not spoof or falsely represent a personal sender number. The selected implementation must comply with applicable Indian SMS and telecom requirements.

Message status and history

- Draft; awaiting approval; scheduled; queued; sent; delivered where provider status is available; failed; retrying; manually resent.

SMS gateway charges and usage limits must be transparent. Do not promise unlimited free SMS.

12. Staff management

- Teacher and staff profiles, contact details and documents.
- Roles, departments, qualifications and subjects taught.
- Assigned batches and classes.
- Staff leave requests and admin approval/rejection.
- Custom permissions for authorised responsibilities.

A salary or payroll module is not required.

13. Homework and study materials

Homework

- Teachers create and publish homework.
- Parents view homework for linked students.
- Support batch-wise and subject-wise organisation.
- Maintain homework history.
- Online student homework submissions are not required.

Study materials

- Teachers upload PDFs and documents.
- Organise materials by batch and subject.
- Parents view and download materials for linked students.
- Apply file permissions and storage limits.

14. Examinations and results

- Create examinations and assign them to batches.
- Teachers manually enter marks.

- Calculate and prepare results.
- Teachers or authorised admins publish results.
- Automatically notify parents when results are published.
- Maintain examination history and individual/batch performance reports.

15. Timetable and academic calendar

Timetable

- Class schedules, subjects, teachers and rooms.
- Breaks, recurring classes and holidays.
- Teacher schedules and room allocation.
- Only admins and authorised staff create or modify timetables.
- Notify parents when relevant timetable changes are published.

Academic calendar

- Holidays, examinations, events, assignments, meetings and academic schedules.
- Admins and authorised staff create and manage calendar events.

16. Communication, enquiries and feedback

- Announcements, circulars, direct messages and group messages.
- Parent-teacher communication through the app.
- Parents submit queries and teachers respond through the app.
- Announcements can include PDF and document attachments.
- Emergency announcements and communication history.
- Basic enquiry management with manually recorded prospective-student details.
- Students and parents can submit feedback or complaints to administrators.
- Only authorised staff can publish announcements.

17. Reports, analytics and global search

- Academic and financial reports.
- Individual student progress and attendance reports.
- Fee collection and pending-payment reports.
- Teacher performance and batch/branch comparisons.
- Downloadable reports.
- Global search across students, staff, parents, courses, batches, fees, announcements and documents.
- Role-based activity feed for announcements, attendance updates, fee records, results, homework and admin activity.

18. AI assistant

EduMan AI is an integrated assistant that understands the current user's role and permissions.

- Draft and personalise SMS reminders.
- Summarise attendance and fee reports.
- Identify attendance patterns and outstanding-fee trends.
- Draft notices and announcements.
- Assist with homework preparation.
- Explain how to use EduMan.
- Help authorised users generate and interpret reports.

AI must not send unapproved messages, publish results without permission, change financial records without authorised confirmation, bypass role permissions or access another institute's private data. Provide limited free AI usage with optional higher limits.

19. Security, audit and recovery

Authentication and access

- Common login page for all organisations.
- Login using organisation code, username and password.
- Secure password hashing and account recovery using SMS OTP or an institute-admin reset request.
- Two-factor authentication for owners and admins.
- Session management and secure logout.
- Role-based access control and server-side authorisation.
- Strict tenant isolation across all institute data.

Audit logs and alerts

- Login history and suspicious activity alerts.
- Profile changes, fee edits and attendance changes.
- Admin actions and permission changes.
- Downloadable audit reports.

Backups and recovery

- Automatic daily backups.
- Manual backups initiated by authorised admins.
- Retain daily backups for 30 days.
- 30-day recycle bin with restoration access for authorised admins.
- Secure restoration and recovery testing.

20. Super Admin and subscriptions

- View and manage registered organisations and users.
- Manage subscription plans, usage limits and subscription status.
- View platform analytics and handle complaints.
- Manage system-wide settings and verification.
- Organisations can request plan upgrades; Super Admin manually approves them.
- Immediate registration with optional verification or approval.
- INR currency for subscription pricing.

Subscription approach

Essential functionality should be free or affordable for small tutors and institutes. Optional premium plans can provide advanced analytics, increased storage, higher AI usage and larger organisational limits. SMS, hosting, storage and AI may have separate usage costs; disclose these clearly.

21. User experience and accessibility

- Professional, information-rich but uncluttered interface.
- Simple workflows suitable for non-technical users.
- Responsive layouts for mobile, tablet and desktop.
- Bottom navigation for main mobile sections.
- Fixed role-specific dashboards.
- Independent light/dark preference for every user.
- Carbon-black dark mode.
- Search, sorting, filters, pagination and column customisation in tables.
- Limited offline access to previously loaded information and selected queued actions.
- Clear loading, success, error and empty states.
- Accessible contrast, keyboard navigation and readable typography.

22. Technical architecture requirements

Use a maintainable architecture with a shared backend and secure APIs for all supported clients. Select technologies only after evaluating cost, reliability, security, deployment and long-term maintainability.

Layer	Responsibilities
Frontend	Responsive web application and shared UI components
PWA and desktop	Installable web app and downloadable Windows/macOS/Linux clients
Android client	Mobile application using the shared backend
Backend/API	Business logic, validation, authorisation and integrations
Database	Relational multi-tenant data with tenant-scoped access
Authentication	Identity, sessions, password recovery and 2FA
File storage	Logos, documents, PDFs and permitted uploads
Background jobs	Scheduled reminders, notifications, reports and retries
AI service layer	Draft generation, summaries and permission-aware assistance
SMS integration	Compliant SMS provider or approved sending mechanism
Monitoring	Logs, errors, metrics and delivery status

Reliability requirements

- Idempotent scheduled jobs and message sending to prevent duplicates.
- Retry handling with limits and failure visibility.
- Server-side validation and permission checks.
- API versioning and documented interfaces.
- Secure file access and upload validation.
- Database migrations and backup restoration tests.
- Monitoring for failed jobs, SMS delivery and application errors.

23. Core data model

Entity	Purpose
Organisations and Branches	Institute workspaces and branches
Users, Roles and Permissions	Accounts, role definitions and access rules
Students and Parents	Student profiles and parent accounts
ParentStudentLinks	Parent-to-student relationships
AcademicYears, Courses and Subjects	Academic structure
Batches and StudentEnrollments	Class groups and student assignments
Attendance and LeaveRequests	Attendance and leave workflows
FeeStructures, StudentFees and Payments	Fee obligations and externally collected payment records
Receipts	Payment receipt records
Examinations and Marks	Exams, marks and published results
Homework and StudyMaterials	Academic tasks and learning resources
Timetables and CalendarEvents	Schedules and academic events
Announcements, Messages and Notifications	Communication and in-app alerts
MessageTemplates, SMSMessages and MessageApprovals	SMS drafts, sending records and approval history
Documents	Uploaded files and metadata
AuditLogs	Security and activity history
Subscriptions and AIUsage	Plan limits and AI usage records

Every organisation-owned record must be associated with the correct organisation. Database constraints, API authorisation and tests must prevent cross-tenant access.

24. Quality assurance and acceptance criteria

- Test registration and multiple institute workspaces under one owner.
- Verify institute records remain isolated, including through direct API requests.
- Test every role and custom permission, including denied access.
- Verify parent-only student-facing login and linked-child visibility.
- Test all-present attendance, absent marking, review and finalisation.
- Verify fee calculations, partial payments, receipts and reminder suppression after full payment.
- Test reminders one day before due date, editing, approval and recipient selection.
- Verify SMS failures, retries, delivery status and duplicate prevention.
- Test in-app notifications independently of SMS delivery.
- Test dashboards, tables, filters, exports and responsive layouts.
- Test backups, restore procedures, recycle-bin retention and audit records.
- Test light mode and carbon-black dark mode.
- Test accessibility, performance, error handling and security.

A feature is not complete merely because its screen exists. It is complete only when its backend logic, permissions, data persistence, error handling and relevant tests work.

25. Product constraints and exclusions

Requirement	Decision
In-app payment gateway	Not included
WhatsApp automated messaging	Not included
Student individual login	Not included
Student homework submission	Not included
Salary and payroll	Not included
Expense management	Not included
Library management	Not included
Transport management	Not included
Inventory management	Not included
Certificate generation	Not included
SMS	Included, subject to compliant provider/device arrangement and charges
AI	Limited free usage with optional higher limits

26. Instructions for the AI app builder

Treat this document as the primary product requirements specification for EduMan. Preserve all stated workflows, constraints, branding decisions and access rules.

- Build real application functionality, not a static mock-up or superficial demo.
- Design the tenant model, database schema, authentication and permissions before implementing business modules.
- Prepare the technical architecture, database schema, role-permission matrix, navigation map, API specification, SMS architecture, security design, task breakdown and test plan for review.
- Use reusable components and maintainable code.
- Do not use mock data as a substitute for production backend functionality.
- Clearly report which requirements are complete, partial or not implemented.
- Test every module and cross-tenant access boundary.
- Do not silently remove, weaken or reinterpret requirements.
- Do not add paid services without explaining costs and obtaining approval.
- Preserve the EduMan name, registration-book logo direction, supplied palette, typography and carbon-black dark mode.
- Do not begin full implementation until the proposed architecture and data model have been reviewed and approved.

The builder must identify technical limitations, external service dependencies, legal or telecom compliance requirements and recurring costs before implementation decisions are finalised.

27. Product summary

EduMan — Manage Every Student. One platform for tutors, tuition centres, coaching centres and educational organisations to manage students, parents, attendance, fees, academics, staff, communication and reporting.

Its defining strengths are simple operation, multi-institute management, parent-centred student access, teacher-friendly attendance, fee tracking, AI-assisted SMS reminders, in-app communication, strong data isolation and affordable access for small organisations.

The central architectural principle is one secure, multi-tenant platform with shared services and consistent experiences across web, mobile and desktop.